Module 07: Communication in Robotics

Learning Objectives

1. Define **Communication** within the context of Robotics.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In Robotics, we see Communication manifest in: * **Specific Example 1**: A distributed Kalman filter across a formation of three drones shares state estimate covariance matrices over a wireless mesh network, where each drone treats its neighbors' transmitted state estimates as additional measurement updates in its own filter; the communication bandwidth directly constrains the precision of shared information -- when packet loss increases, each drone's filter automatically downweights the missing neighbor's contribution (reducing that channel's precision), demonstrating how Active Inference's precision-weighting naturally handles unreliable communication channels in multi-robot estimation. * **Specific Example 2**: A CAN-bus communication backbone on an industrial robot transmits joint encoder readings, motor current measurements, and torque commands between the central controller and distributed motor drives at 1 ms intervals; timing jitter and bus congestion introduce variable delays that the state estimator must account for by adjusting the prediction horizon of its generative model -- messages arriving late carry stale information with higher uncertainty, and the controller's communication protocol implicitly encodes this as reduced precision on delayed observations.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.